



Department of Mathematical & Physical Sciences

EAST WEST UNIVERSITY

Aftabnagar, Dhaka 1212, Bangladesh

Course Outline of Course PHY 109 (Engineering Physics-I)

Semester: Summer 2025; Sections: 6;

Course Instructor: Md. Mahfuzul Haque (MMHQ)

Course Information

Course Code : PHY 109
Course Title : Engineering Physics – I
Pre-requisite : MAT 205
Credit Hours : 3
Class Time : **Section** **Day** **Time** **Room No.**
 1 ST 08.30AM-10.00AM FUB-603

Office Hour :

Days	Office Hours
Sun (S)	10.00AM-03.00PM
Mon (M)	03.00PM-04.50PM
Tues (T)	10.00AM-11.50AM 03.00PM-04.00PM
Wed (W)	03.00PM-04.50PM
Thurs (R)	03.00PM-04.00PM

Make-up Schedule : Makeup classes will be suitably arranged.

Instructor Information

Course Instructor : Md. Mahfuzul Haque
E-mail : mahfuzul.haque@ewubd.edu
Office : Room # 705, Building AB1

Course Objective

Physics is the subject that deals about the principles of laws of matter and energy. To build up a career in engineering discipline it is very important to have knowledge of Physics. Physics is the basic/fundamental subject in science and engineering. Physics plays always important role in developing of new technology. The course Engineering Physics I has been designed to give a clear concept in the area of mechanics, fluid dynamics, SHM, waves (mechanical and electromagnetic), thermal processes and optics. All these areas are basic/fundamental in Physics. The concept in these areas can help the students for studying different engineering and communication courses.

Learning Outcomes

After completing the course, the students will be able to:

1. Apply the concept of force in their higher courses.
2. Understand the fluid flow and concepts of Continuity and Bernoulli's equations
3. Know the different types of waves and how wave convey energy.
4. Know thermodynamic laws and entropy and ideal heat engine.
5. Know about light and interference of light, diffraction of light and polarization of light.
6. Know about the concept of electric and magnetic field and different laws related to electro-magnetism.

Course Contents/Descriptions

1. Mechanics: Newton's law of motion; Kinetic energy, potential energy and work; Momentum, impulse, Collisions; Rotation of rigid body; Rotational dynamics; Equilibrium of rigid bodies.
2. Fluid Mechanics: Concept of fluids; Pressure and density; Measurement of pressure; General concept of fluid flow; The equation of continuity; Bernoulli's equation; Applications.
3. Oscillations and Waves: Simple harmonic oscillation, energy in SHM, damped harmonic motion; Different types of waves; The superposition principle; Wave speed; Energy and Power of traveling wave along string; Interference of waves; standing waves and resonance.
4. Thermal Physics: Concept of temperature and heat; isothermal and adiabatic changes; Reversible and irreversible processes; Three laws of thermodynamics; Thermodynamics and the concept of entropy, Heat engine.
5. Electricity and Magnetism: Concept of electric charge, the electric field, dipole in an electric field, electric flux, Gauss' law, electric potential, calculation of capacitance, capacitors with dielectric, energy storage in an electric field. The magnetic field, Biot-Savart law, Ampere's law, magnetic force on a current, magnetic lines of induction, Faraday's law, Lenz's Law.
6. Physical Optics: Light as electromagnetic wave; interference, diffraction of light, polarization of light; Young's double slit experiment; Michelson interferometer, Poynting vector.

Text Book

1. Fundamentals of Physics; Halliday, Resnick and Walker, 10th Edition, Wiley & Sons.

Reference books:

1. University Physics; Sears and Zemansky's, Hugh D. Young, Roger A. Freedman, Pearson Education, Inc. London.
2. Mechanics, Wave Motion & Heat; Sears, F.W.; Addison Wesley Publishing Company
3. Electricity and Magnetism, Dakworth, H. E.
4. Introduction to Optics, By Frank L. Pedrotti, Leno M. Pedrotti, Leno S. Pedrott

Assessment Tools:

Assessment tools include **Class Tests (Short Quizzes)**, **Presentations/Assignment**, and **Exams**. The Class Tests are about 20-30 minutes in class and the Mid Term Exam and Final Exam are 60 - 80 minutes duration in class. Class Test and Exam dates will be announced in the Class.

Assessment/Evaluation/Grading Policy:

The relative contributions of class tests, presentations and exams are as follows:

Test/Exam	% of Marks
Class Tests (Quiz)	20
Assignment/Presentation	10
Mid Term Exam	30
Final Exam	40

The University Grading Scheme is the following

Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point
80 – 100	A+	4.00	65 – 69	B+	3.25	50 – 54	C+	2.50
75 – 79	A	3.75	60 – 64	B	3.00	45 – 49	C	2.25
70 – 74	A-	3.50	55 – 59	B-	2.75	40 – 44	D	2.00
						Less than 40	F	0.00

Essential Policy Information

1. There is zero tolerance for cheating at EWU. Students caught with cheat sheets in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of hand, back of calculators, chairs or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.

2. Regular Class attendance will be taken. The students are advised to attend classes regularly.
3. Makeup exam for mid terms and final may be arranged for compelling reasons. For the makeup exam, the student must apply with supporting documents.



(Md. Mahfuzul Haque)

Course Instructor